OPPORTUNITY BRIEF · SALES AND PRESALES

Federated learning for privacy-preserving anomaly detection in industrial IoT networks to secure smart manufacturing operations.

Manufacturing × OT and industrial control system security × Machine learning · OS084 v1 · refreshed 2026-08-18

ATTRACTIVENESS	RIGHT TO WIN	COMPETITION	SERVICEABLE MARKET	HORIZON	PORTFOLIO
47 is the world moving	**74** can we play, can we win	**MEDIUM** 0 named in the evidence	**€309m** observed evidence · per year	**LATER** research and standards activi...	**L1** 18 signals

Attractiveness and right to win are scored and displayed separately and are never combined (SC-12). Competition is a fourth quantity, not a discount on either.

The opportunity

Federated learning for privacy-preserving anomaly detection in industrial IoT networks offers manufacturers a way to secure smart manufacturing operations without exposing sensitive production data. This opportunity is for manufacturing firms with distributed OT/ICS environments that need to detect anomalies across sites while respecting data privacy. It exists now because recent research demonstrates viable federated learning frameworks that combine anomaly detection with privacy-preserving techniques like homomorphic encryption and differential privacy.

Market opportunity

Method	Total (TAM)	Serviceable (SAM)	Obtainable (SOM)	Confidence
Bottom-up: enterprises x adoption x contract value	**€431m** €77.6m – €2.3bn	**€309m** €55.7m – €1.7bn	**€14.9m** €2.7m – €79.8m	observed
Observed: tendered contract value, annualised	**€20.9m** €20.9m – €56.6m	**€20.9m** €20.9m – €56.6m	**€1.0m** €1.0m – €2.7m	observed

All figures are annual, in EUR. Ranges compound the uncertainty of each factor below; the base case is the middle column of each.

Bottom-up: enterprises x adoption x contract value

Factor	Value	Basis	Source	As of
Enterprises in Manufacturing (EU27_2020)	238,122 enterprises	observed	Eurostat · sbs_sc_ovw	2024
Enterprises adopting Machine learning	3.3 % of enterprises (10+ employees)	observed	Eurostat · isoc_eb_ain2 · E_AI_TML	2025
Annual value of one engagement	€337k per enterprise per year	observed	TED (Tenders Electronic Daily) · ted	2026-05-19..2026-08-05
Size-weighted buyer base	39,220 enterprise-equivalents at large-enterprise engagement value	assumption	config/sizing.yaml · size_class_value_weights	size-2026-08-a
Assumed obtainable share of the serviceable market	4.8 % of SAM	assumption	config/sizing.yaml · obtainable_share	size-2026-08-a

Read this before quoting the number. Contract values are observed from EU public procurement (TED). For private-sector verticals they are used as a proxy for comparable enterprise deal size: the buying entity differs, the scope of work is comparable. Engagement value is scaled by enterprise size class (10-19: x0.04, 20-49: x0.09, 50-249: x0.35, GE250: x1.0), because the observed contract value comes from large-organisation tenders.

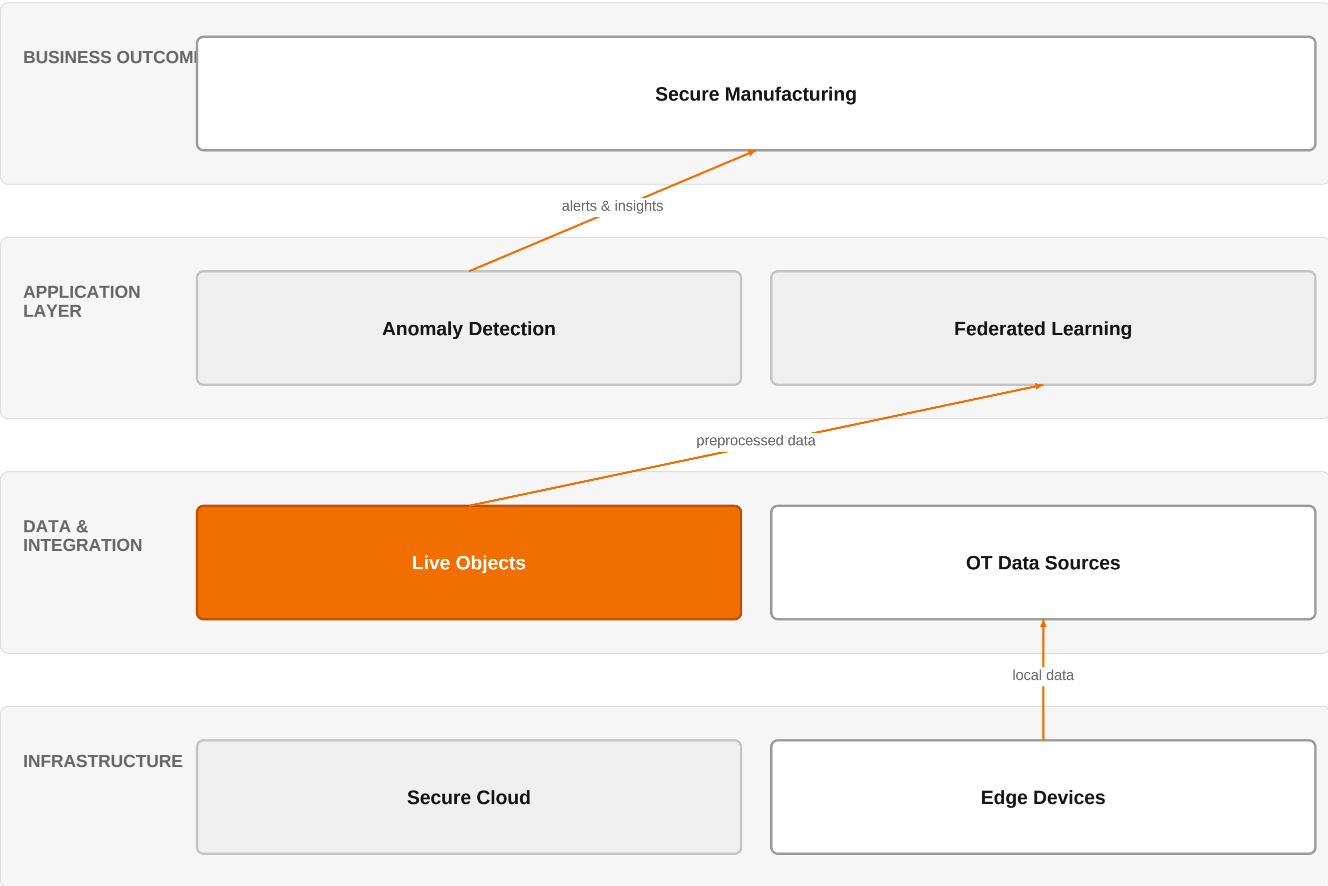
Observed: tendered contract value, annualised

Factor	Value	Basis	Source	As of
Tendered contract value in matching CPV categories	€20.9m per year	observed	TED (Tenders Electronic Daily) · ted	2026-05-19..2026-08-05
Assumed obtainable share of the serviceable market	4.8 % of SAM	assumption	config/sizing.yaml · obtainable_share	size-2026-08-a

Read this before quoting the number. Observed over 85 days of TED coverage and scaled to a year (x4.3); a short collection window makes that scaling rough. 14 matching notices, matched on vertical via the CPV crosswalk — §4.5.2's warning applies: a crosswalk error lands directly in this number. Public procurement only. Private-sector demand for the same capability is not visible here at all, so this is a floor rather than a market size.

The solution, in one picture

Federated Anomaly Detection for Industrial IoT



Orange asset · Partner · Customer-owned · Third party / to be sourced

This diagram shows how Orange's federated learning solution detects anomalies across industrial IoT sites without centralizing sensitive data, ensuring privacy and security.

Boxes marked as Orange assets are validated against the linked business graph; a box the model claimed for Orange without a matching asset is shown as third party.

What has changed

Industrial IoT adoption is increasing, introducing significant cybersecurity and privacy challenges, particularly in anomaly detection and secure data sharing. Traditional centralized intrusion detection systems face limitations in latency, scalability, and data privacy, while signature-based systems cannot adapt to zero-day attacks. Federated learning offers a privacy-preserving approach, with recent frameworks integrating adaptive anomaly detection, homomorphic encryption, and differential privacy. Additionally, industrial control systems face dual risks in cyber and physical domains, and current risk assessment methods often fail to consider cross-domain propagation, highlighting the need for more integrated security approaches.

`Sources: SIG-F633044AF87A openalex.org 2026-01-29; SIG-CEE282CB905A openalex.org 2025-09-08; SIG-8EC0D797B8D2 openalex.org 2025-10-27; SIG-005CE36DA1FC Elsevier BV 2026-01-01; SIG-FEA6096A63EB Elsevier BV 2026-01-01`

Who buys, and why now

The Chief Information Security Officer (CISO) or OT Security Manager typically signs, while the plant operations manager feels the pain of potential production downtime due to undetected anomalies. The trigger is often a security incident or audit finding that reveals gaps in detecting anomalies across distributed sites without sharing sensitive process data. They need to demonstrate compliance with data privacy regulations while improving threat detection across their industrial IoT estate.

Why it is hot now — every claim bound to a dated source

- Privacy-preserving federated learning frameworks for anomaly detection in smart industrial networks address security and privacy challenges. [SIG-F633044AF87A]

- Federated learning approaches are proposed for reliable anomaly detection under clustered data distribution in industrial IoT. [SIG-1537508DCBEE]
- A secure federated learning framework integrates adaptive anomaly detection with homomorphic encryption and differential privacy for IIoT. [SIG-F633044AF87A]
- Decentralized federated learning defends blockchain-based industrial IoT systems against Sybil attacks. [SIG-6B094A3F3A46]
- Federated learning in industrial IoT addresses clustered data distributions for reliability. [SIG-1537508DCBEE]
- Industrial control systems face dual risks in cyber and physical domains, and current risk assessments often fail to consider cross-domain propagation. [SIG-005CE36DA1FC, SIG-FEA6096A63EB]
- There is an urgent need for research on risk assessment that combines safety and security, as purely safety or security assessments are insufficient. [SIG-FEA6096A63EB]
- Intrusion detection in industrial control systems can be enhanced using decision tree based invariants. [SIG-B3004649A8C9]

What Orange would deliver, and why it can win

What Orange would deliver

Orange would assemble a federated learning-based anomaly detection solution using its AI/Data/Cloud experts, Cybersecurity experts, and IoT experts. The engagement would start with a discovery phase using Orange's IoT expertise to map the customer's industrial IoT architecture and data flows. Then, Orange would deploy a federated learning framework, leveraging its cybersecurity expertise to ensure privacy-preserving techniques like homomorphic encryption and differential privacy are applied. The solution would be delivered on Orange's secure infrastructure, with optional integration of Orange Cyberdefense managed services for ongoing monitoring. Orange's Live Objects platform could serve as the IoT data ingestion layer, while partnerships with cloud providers like AWS or Google Cloud could provide scalable compute resources.

Why Orange can win

Orange can win by combining its deep expertise in AI, cybersecurity, and IoT with its ability to deliver sovereign, certified solutions (ISO 27001, TISAX). The customer's trust in Orange's managed services and its experience with large industrial clients (e.g., Colgate-Palmolive, Mondelez International) provide credibility. However, the specific federated learning technology is not a named Orange asset, so Orange would need to partner or build on research from Orange Group researchers to differentiate.

Named assets behind this — each individually inspectable (LK-08)

Asset	Type	Link	Why it is linked	Curator
Live Objects	offer	L1	offer provides the technology in a matching domain	unconfirmed
Orange Drone Guardian	offer	L1	offer provides the technology in a matching domain	unconfirmed
Orange Cyberdefense managed services	offer	L1	offer addresses the use case but not this technology	unconfirmed
AWS	partner	L2	partner provides the required technology	unconfirmed
Google Cloud	partner	L2	partner provides the required technology	unconfirmed
NiCE	partner	L2	partner provides the required technology	unconfirmed
NVIDIA	partner	L2	partner provides the required technology	unconfirmed
ISO 27001	certification	SUP	certification Orange holds is required by this vertical	unconfirmed
TISAX	certification	SUP	certification Orange holds is required by this vertical	unconfirmed
AI / Data / Cloud experts	capability pool	SUP	capability pool staffs this domain, vertical or technology	unconfirmed
Cybersecurity experts	capability pool	SUP	capability pool staffs this domain, vertical or technology	unconfirmed
IoT experts	capability pool	SUP	capability pool staffs this domain, vertical or technology	unconfirmed
Defense and security experts	capability pool	SUP	capability pool staffs this domain, vertical or technology	unconfirmed

Orange Group researchers	capability pool	SUP	capability pool staffs this domain, vertical or technology	unconfirmed
Colgate-Palmolive	reference	SUP	published reference in this vertical, different use case	unconfirmed
Mondelez International	reference	SUP	published reference in this vertical, different use case	unconfirmed

6 further published reference(s) in this vertical are linked to this topic and are listed in the radar.

Competitive landscape

MEDIUM competitive intensity (score 7.1; 0 of 8 listed players appear in this topic's own evidence)

A real field, including at least one player the customer will already know. Expect to be compared.

Competitor	Category	Position here	Basis	Evidence
AWS	Hyperscaler	sells Machine learning; competes in OX: Smart Industries — also an Orange partner	structural	—
Google Cloud	Hyperscaler	sells Machine learning — also an Orange partner	structural	—
Atos / Eviden	Global systems integrator	sells Machine learning; competes in Cybersecurity	structural	—
NiCE	Customer-experience platform	sells Machine learning — also an Orange partner	structural	—
Databricks	Data and AI platform	sells Machine learning	structural	—
Dataiku	Data and AI platform	sells Machine learning	structural	—
Palantir	Data and AI platform	sells Machine learning; competes in OX: Smart Industries	structural	—
Accenture	Global systems integrator	active in Manufacturing; competes in OX: Smart Industries	structural	—

evidenced means this topic's own sources name the competitor — the signal ids are in the last column and are dated and clickable in the radar. **structural** means the curated register says they sell this technology into this vertical, which is true but is not proof they are in this deal. Register version competitors-2026-08-a.

Qualifying questions for the first meeting

1. Do you have multiple industrial sites or OT networks that need to share threat intelligence without exposing sensitive process data?
2. Have you experienced a security incident or audit finding that highlighted gaps in anomaly detection across your IIoT environment?
3. Are you currently using centralized intrusion detection systems that struggle with data privacy or latency?
4. Do you have a data governance policy that restricts sharing raw sensor data across sites or with third parties?
5. Is your OT security team under pressure to improve detection of zero-day attacks or unknown anomalies?
6. Are you exploring federated learning or other privacy-preserving AI techniques as part of your security roadmap?

Objections you will meet

They will say	You can answer
Federated learning is too immature for our production OT environment.	You're right that federated learning is still emerging, but recent research shows practical frameworks with adaptive anomaly detection and privacy preservation. We would start with a pilot on non-critical systems to prove value and build confidence.
We already have security solutions from our cloud provider.	Cloud providers offer strong ML platforms, but they may not have the OT-specific expertise or the privacy-preserving features needed for industrial environments. Orange brings deep OT knowledge and can integrate with your existing cloud investments while adding a managed service layer.
We are concerned about sharing data with a third party.	That's exactly why federated learning is relevant—it trains models without sharing raw data. Orange can deploy the solution on your premises or in a sovereign cloud, and we hold ISO 27001 and TISAX certifications to ensure data protection.

Risks and unknowns

The main risk is that federated learning for anomaly detection is still largely research-oriented, with limited production deployments. The technology may not yet be mature enough for mission-critical OT environments. Additionally, the integration with existing ICS and the handling of clustered data distributions remain challenges. There is also uncertainty about regulatory acceptance of federated learning for compliance purposes. Orange's ability to deliver this depends on its research capabilities and partnerships, which are not fully proven in this specific area.

Next action, by role (FR-17)

Role	Do this next
Presales / Proposal	Prepare a bid brief for a manufacturing client's OT security proposal, incorporating federated learning concepts and highlighting our Live Objects IoT platform and Orange Cyberdefense managed services, with TISAX certification as a differentiator.
Sales	In a conversation with a manufacturing account's OT security lead, mention that Orange Cyberdefense managed services can help them explore privacy-preserving anomaly detection approaches for their industrial IoT networks, referencing our ISO 27001 certification.
Strategist / Innovator	Commission a deep-dive brief on federated learning for privacy-preserving anomaly detection in industrial IoT, leveraging Orange Group researchers and IoT experts to map the technology's maturity against our AI/Data/Cloud and Cybersecurity expert pools.

Evidence

Every claim in this brief resolves to one of these. Tier 1 is an authoritative primary source; tier 4 is vendor-published material, discounted in scoring (§4.3.7).

Signal	Date	Publisher	T	Title and link
SIG-D2558A0D8EDD	2026-08-01	Computer Communications	1	Unsupervised CoAP-based anomaly detection in private 5G core network for Industrial-IoT
SIG-C6ED90595BBC	2026-04-18	openalex.org	1	Towards collaborative and privacy-preserving industrial time-series anomaly detection with fede...
SIG-6B094A3F3A46	2026-03-26	openalex.org	1	Sybil attack defense in blockchain-based industrial IoT systems using decentralized federated l...
SIG-1537508DCBEE	2026-03-10	openalex.org	1	FedCST: Reliable federated learning under clustered data distribution in industrial internet of...
SIG-F633044AF87A	2026-01-29	openalex.org	1	SecuFL-IoT: an adaptive privacy-preserving federated learning framework for anomaly detection i...
SIG-005CE36DA1FC	2026-01-01	Elsevier BV	1	A quantitative risk assessment method for industrial control system safety and security
SIG-FEA6096A63EB	2026-01-01	Elsevier BV	1	A Survey on Key Issues of Risk Assessment methods for Industrial Control System Security and Sa...
SIG-AECDD651DA98	2025-12-17	openalex.org	1	Edge-AI integrated secure wireless IoT architecture for real time healthcare monitoring and fed...
SIG-904D344AF0C0	2025-10-28	openalex.org	1	Federated learning based on two-stage knowledge distillation for intrusion detection in industr...
SIG-C8282154F431	2025-10-28	openalex.org	1	Lightweight Signal Processing and Edge AI for Real-Time Anomaly Detection in IoT Sensor Networks
SIG-8EC0D797B8D2	2025-10-27	openalex.org	1	AI-Powered Security for IoT Ecosystems: A Hybrid Deep Learning Approach to Anomaly Detection
SIG-98BCC6F19BE4	2025-10-27	openalex.org	1	QuantumTrust-FedChain: A Blockchain-Aware Quantum-Tuned Federated Learning System for Cyber-Res...
SIG-A49A291BF70B	2025-10-01	openalex.org	1	PureChain-enhanced federated learning for dynamic fault tolerance and attack detection in distr...
SIG-CF12E7BF5186	2025-09-22	openalex.org	1	ROCHE: A Robust and End-to-End Privacy-Preserving Federated Learning Framework for Intrusion De...
SIG-B75BEF75986F	2025-09-08	openalex.org	1	Trustworthy Adaptive AI for Real-Time Intrusion Detection in Industrial IoT Security
SIG-CEE282CB905A	2025-09-08	openalex.org	1	Survey of Federated Learning for Cyber Threat Intelligence in Industrial IoT: Techniques, Appli...
SIG-B3004649A8C9	2025-09-01	Computers & Security	1	Decision tree based invariants for intrusion detection in industrial control system
SIG-8FD8EE65E437	2025-08-19	openalex.org	1	Differential privacy-enabled federated learning for secure neural synchronization in protecting...

How this brief was made

Computed, not written. Attractiveness, right to win, competitive intensity and every figure in the market-opportunity section are computed from stored inputs and can be re-derived (DR-05, NFR-01). No language model produced a number anywhere in this document (§4.4.4).

Written, then checked. The narrative sections and the solution diagram were generated from this topic's evidence and from the named asset and competitor lists. Factual sections must cite signal ids attached to this topic; sections that failed that check, or that contained a generated quantity or an unsupplied organisation, were removed rather than rewritten.

Removed by those checks in this brief: competitive_landscape (no claim survived evidence binding); diagram (2 box(es) claimed to be an Orange asset that was not supplied — shown as third-party instead)

What	Version	When
Scores — weight set	w-2026-08-a	2026-08-18T19:08:26+00:00
Market sizing — assumptions	size-2026-08-a	2026-08-18T21:06:15+00:00
Competitor register	competitors-2026-08-a	2026-08-18T21:06:19+00:00
Narrative — prompt / model	describe-v1 / deepseek-chat	2026-08-19T04:56:43+00:00
Pipeline	0.1.0	2026-08-18

Generated 2026-08-19 04:58 UTC. Scores computed under a different weight set are not comparable with these (SC-10), and neither are market sizes computed under a different sizing version.